

LUCAS PESENTI

POSITIONS AND EDUCATION

Scientific assistant and postdoctoral researcher

Sep. 2025 – Aug. 2026

ETH Zürich

- Hosted by David Steurer

Ph.D. in Statistics and Computer Science

Sep. 2021 – Jan. 2026

Bocconi University, Milan

- Thesis: “Algorithms beyond the union bound: polynomial optimization and discrepancy theory”
- Advised by Luca Trevisan (until June 2024), Laura Sanità and Pravesh Kothari (since June 2024).
- Awarded cum laude on January 21st, 2026.

B.Sc. and M.Sc. in Computer Science

Sep. 2016 – Sep. 2021

École Normale Supérieure de Paris (ENS Ulm)

- Parisian Master of Research in Computer Science obtained with the highest honors (rank 1/65).
- Student and civil servant at ENS with a major in computer science and a minor in mathematics.
- Ranked 1st at the ENS entrance exam (computer science section).

MANUSCRIPTS

- [1] *Norm bounds for sparse random tensors and spectral gap of random hypergraphs*, with Kevin Lucca, 2026.
- [2] *Universality of first-order methods on random and deterministic matrices*, with Nicola Gorini, Dmitriy Kunisky and Chris Jones, 2026.
- [3] *Agnostic learning in (almost) optimal time via Gaussian surface area*, with Lucas Slot and Manuel Wiedmer, 2026.

PUBLICATIONS

- [4] *Fourier analysis of iterative algorithms*, with Chris Jones, 2025.
Proceedings of the International Colloquium on Automata, Languages and Programming (ICALP).
- [5] *New SDP roundings and certifiable approximation for cubic optimization*, with Jun-Ting Hsieh, Pravesh K. Kothari, and Luca Trevisan, 2024.
Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms (SODA).
- [6] *Discrepancy minimization via regularization*, with Adrian Vladu, 2023.
Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms (SODA).

TEACHING ASSISTANTSHIP

(ETH Zürich) Algorithms and Data Structures (Head TA)	Fall 2025
(Bocconi University) 20874 – Algorithms for Optimization and Inference	Fall 2023, Fall 2024
(Bocconi University) 20873 – Analysis of Algorithms and Data Structures	Fall 2023, Fall 2024

SELECTED COMMUNICATIONS

<i>Fourier analysis of iterative algorithms</i> International Colloquium on Automata, Languages and Programming (ICALP 2025), Aarhus.	Jul. 2025
<i>Fourier analysis of random optimization problems</i> Johns Hopkins University.	Apr. 2025
<i>On maximizing polynomials</i> Workshop in memory of Luca Trevisan (1971-2024), Bocconi University.	Dec. 2024
<i>Understanding iterative algorithms with Fourier diagrams</i> DACO seminar, ETH Zürich.	Oct. 2024
<i>Diagram analysis of iterative algorithms</i> Rocella Conference on Inference and AI (poster presentation).	Sep. 2024
<i>Up to the constant: Spencer's theorem and random matrices</i> Computer science seminar, University of Pennsylvania.	Feb. 2024
<i>New SDP roundings and certifiable approximation for cubic optimization</i> Symposium on Discrete Algorithms (SODA 2024), Washington DC.	Jan. 2024
<i>Discrepancy minimization via regularization</i> Symposium on Discrete Algorithms (SODA 2023), Florence.	Jan. 2023

SERVICE

Reviewing for Symposium on Foundations of Computer Science (FOCS), Symposium on Theory of Computing (STOC), Symposium on Discrete Algorithms (SODA), Conference on Learning Theory (COLT), SIAM Journal on Computing (SICOMP), Journal of the ACM (JACM), Mathematical Statistics and Learning (MSL), Random Structures and Algorithms (RSA).	
Ph.D. representative	2022 – 2025
Organization of the theoretical computer science reading group	2021 – 2023

EXTRA-ACADEMIC ACHIEVEMENTS

Bronze medals in the ACM-ICPC Southwestern European contest.	2016, 2017
Bronze medal in the International Olympiad in Informatics (IOI, Taipei)	2014
Ranked 1st in the national IOI selection	2013 – 2014